EXCELLENCE

CLASS 11 - 2026-27

Physics | Lecture - 17

Basic Maths and Calculus

Integration by Substitution and Mixed Practice

By - Excellence Programme

24 September 2026

Topics to be covered

01 5-minute definite-integral recap

02 Substitution idea: chain rule backwards

03 Choose u and write du

04 Easy indefinite substitutions

05 Definite integrals with substitution

06 Change of limits

07 Mixed board practice

08 Physics applications

Definite integrals from last class

$\int_a^b f(x)\,dx = F(b) - F(a)$ (no +C)

If $f \geq 0$, this is the area under the curve

Displacement = area under the v–t graph

$\int x^n\,dx = x^{n+1}/(n+1) + C$ $(n \neq -1)$

Quick check: definite

Evaluate ∫_0Â³ 2x dx.

Solution: Quick check: definite

Step 1: $F(x) = x^2$

Step 2: $F(3) - F(0) = 9 - 0$

9.

Rough work / extra work

Substitution: chain rule in reverse

Chain rule: d/dx [F(u)] = F'(u) · du/dx.

So if the integrand looks like f(u) · (du/dx), integrate with respect to u.

$$\int f(u)\,(du/dx)\,dx = \int f(u)\,du$$

Always replace dx as well — do not leave a mixture of u and x.

Four board steps

1. Choose u = (the inside function)

2. Differentiate: du/dx = … → write du = (…) dx

3. Rewrite the whole integral in u only

4. Integrate, then put x back (or change limits)

Simple linear inside

Find $\int (2x + 3)^3 \cdot 2\,dx$.

Solution: Simple linear inside

Step 1: Let $u = 2x + 3 \rightarrow du = 2\,dx$

Step 2: Integral becomes $\int u^3\,du = u^4/4 + C$

$(2x + 3)^4/4 + C$.

Missing constant factor

Find $\int (2x + 3)^3 \, dx$.

Solution: Missing constant factor

Step 1: $u = 2x + 3 \rightarrow du = 2\,dx \rightarrow dx = du/2$

Step 2: $\int u^3 (du/2) = (1/2) \cdot u^4/4 + C = u^4/8 + C$

$(2x + 3)^4/8 + C.$

If du has a constant factor, bring 1/that factor outside.

Rough work / extra work

Power of a linear expression

Find $\int \sqrt{(3x + 1)}\, dx$.

Solution: Power of a linear expression

Step 1: u = 3x + 1 → du = 3 dx → dx = du/3

Step 2: ∫ uÂ¹á❺ŸÂ² (du/3) = (1/3) · uÂ³á❺ŸÂ²/(3/2) + C

Step 3: = (1/3)·(2/3) uÂ³á❺ŸÂ² + C = (2/9) uÂ³á❺ŸÂ² + C

(2/9)(3x + 1)Â³á❺ŸÂ² + C.

Exponential substitution

Find $\int e^{2x} \cdot 2\,dx$.

Solution: Exponential substitution

Step 1: $u = 2x \rightarrow du = 2\,dx$

Step 2: $\int e^u\,du = e^u + C$

$e^{(2x)} + C$.

Rough work / extra work

Exponential without the 2

Find ∫ e^(2x) dx.

Solution: Exponential without the 2

Step 1: $u = 2x \rightarrow dx = du/2$

Step 2: $\int e^u (du/2) = (1/2) e^u + C$

$(1/2) e^{(2x)} + C$.

Sine of a linear angle

Find $\int \sin(3x)\,dx$.

Solution: Sine of a linear angle

Step 1: $u = 3x \rightarrow dx = du/3$

Step 2: $\int \sin u \, (du/3) = -(1/3)\cos u + C$

$-(1/3)\cos(3x) + C$.

Cosine of a linear angle

Find $\int \cos(5x)\,dx$.

Solution: Cosine of a linear angle

Step 1: $u = 5x \rightarrow dx = du/5$

$(1/5)\sin(5x) + C.$

Rough work / extra work

Chain-rule lookalike

Find $\int 2x \cos(x^2)\, dx$.

Solution: Chain-rule lookalike

Step 1: $u = x^2 \;\rightarrow\; du = 2x\,dx$

Step 2: $\int \cos u\,du = \sin u + C$

$\sin(x^2) + C$.

Reciprocal substitution

Find $\int (2x)/(x^2 + 1)\, dx$.

Solution: Reciprocal substitution

Step 1: $u = x^2 + 1 \rightarrow du = 2x\,dx$

Step 2: $\int du/u = \ln|u| + C$

$\ln|x^2 + 1| + C$.

Rough work / extra work

Definite integrals with substitution

Two clean options: change the limits to u, or substitute back to x then use F(b) − F(a).

Preferred: change limits with u → no need to return to x

If you change variable, you must also change both limits.

Changing limits

When x = a, compute u(a)

When x = b, compute u(b)

Then evaluate ∫_{u(a)}^{u(b)} f(u) du

Do not put x-limits on a u-integral.

Definite with linear u

Evaluate $\int_0^1 (2x + 1)^3 \cdot 2\,dx$.

Solution: Definite with linear u

Step 1: $u = 2x + 1 \rightarrow du = 2\,dx$

Step 2: $x = 0 \rightarrow u = 1$; $x = 1 \rightarrow u = 3$

Step 3: $\int_{1}^{3} u^3\,du = [u^4/4]_{1}^{3} = 81/4 - 1/4$

20.

Definite sine substitution

Evaluate $\int_0^{\pi/6} \cos(3x) \cdot 3\,dx$.

Solution: Definite sine substitution

Step 1: $u = 3x \rightarrow du = 3\,dx$

Step 2: $x = 0 \rightarrow u = 0;\quad x = \pi/6 \rightarrow u = \pi/2$

Step 3: $\int_0^{\pi/2} \cos u\,du = [\sin u]_0^{\pi/2} = 1$

1.

Rough work / extra work

Definite without the factor ready

Evaluate $\int_0^1 e^{(2x)}\,dx$.

Solution: Definite without the factor ready

Step 1: $u = 2x \rightarrow dx = du/2$; $x = 0 \rightarrow u = 0$; $x = 1 \rightarrow u = 2$

Step 2: $\int_{0}^{2} e^u (du/2) = (1/2)[e^u]_{0}^{2} = (1/2)(e^2 - 1)$

$(e^2 - 1)/2$.

Definite reciprocal form

Evaluate $\int_0^1 \frac{2x}{(x^2 + 1)}\,dx$.

Solution: Definite reciprocal form

Step 1: $u = x^2 + 1 \rightarrow du = 2x\,dx$

Step 2: $x = 0 \rightarrow u = 1$; $x = 1 \rightarrow u = 2$

Step 3: $\int_{1}^{2} du/u = [\ln|u|]_{1}^{2} = \ln 2 - \ln 1$

$\ln 2$.

Rough work / extra work

Physics with substitution

When acceleration or velocity is a composite function of time, substitute the inside.

Same four steps — then put the physical limits of t

Keep units on the final number.

Velocity from a = 2t + 1

If a = 2t + 1 and v = 0 at t = 0, find v at t = 2.

Solution: Velocity from a = 2t + 1

Step 1: $v = \int_0^2 (2t + 1)\,dt = [t^2 + t]_0^2$

Step 2: $= (4 + 2) - 0 = 6$

$v = 6\ m\ s^{-1}$.

Displacement with $v = 3t^2$

If $v = 3t^2$ and $x = 0$ at $t = 0$, find x at $t = 2$.

Solution: Displacement with $v = 3t^2$

Step 1: $x = \int_0^2 3t^2\,dt = [t^3]_0^2 = 8$

$x = 8$ m.

Rough work / extra work

Composite velocity

If $v = 2t\cos(t^2)$ and $x = 0$ at $t = 0$, find x at $t = \sqrt{(\pi/2)}$.

Solution: Composite velocity

Step 1: $u = t^2 \;\rightarrow\; du = 2t\,dt;\quad t = 0 \rightarrow u = 0;\quad t = \sqrt{(\pi/2)} \rightarrow u = \pi/2$

Step 2: $x = \int_0^{\pi/2} \cos u\,du = [\sin u]_0^{\pi/2} = 1$

$x = 1$ m.

Mixed: identify u

Find $\int 6x^2 (x^3 + 4)^5 \, dx$.

Solution: Mixed: identify u

Step 1: $u = x^3 + 4 \rightarrow du = 3x^2\,dx \rightarrow 6x^2\,dx = 2\,du$

Step 2: $\int 2u^5\,du = 2 \cdot u^6/6 + C = u^6/3 + C$

$(x^3 + 4)^6/3 + C.$

Mixed: definite power

Evaluate $\int_0^1 4x\,(x^2 + 1)^3\,dx$.

Solution: Mixed: definite power

Step 1: $u = x^2 + 1 \;\to\; du = 2x\,dx \;\to\; 4x\,dx = 2\,du$

Step 2: $x = 0 \to u = 1;\quad x = 1 \to u = 2$

Step 3: $\int_{1}^{2} 2\,u^3\,du = 2[u^4/4]_{1}^{2} = (1/2)(16 - 1)$

15/2.

Rough work / extra work

Homework: sin(2x)

Find $\int \sin(2x)\,dx$.

Solution: Homework: sin(2x)

Step 1: $u = 2x \rightarrow dx = du/2$

$-(1/2)\cos(2x) + C.$

Homework: definite e^(3x)

Evaluate ∫_0^1 e^(3x) dx.

Solution: Homework: definite e^(3x)

Step 1: u = 3x → dx = du/3; limits 0 → 3

Step 2: (1/3)[e^u]_0Â^3 = (e^3 − 1)/3

(e^3 − 1)/3.

Rough work / extra work

Homework: check by differentiate

Differentiate y = (1/5) sin(5x) and confirm it matches cos(5x).

Solution: Homework: check by differentiate

Step 1: $dy/dx = (1/5) \cdot \cos(5x) \cdot 5 = \cos(5x)$

Check successful — so $\int \cos(5x)\,dx = (1/5)\sin(5x) + C$.

Substitution lock lines

Substitution = chain rule written backwards

Choose u, write du, rewrite the whole integral

For definite integrals, change both limits with u

Always check by differentiating the answer

Practice

1. Redo every substitution question without looking.

2. For each indefinite answer, differentiate and recover the integrand.

3. Practise changing limits on three definite examples.

4. Write one physics $\int$ with a composite function of t.

Leaving a mix of u and x in the same integral is the classic trap.